Report 2: The Zak Transform as a Reshape plus an FFT

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S2

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1 What this report is about

- The Zak transform is the tool that moves a time signal into the delay–Doppler (DD) grid of Report 1.
- Despite its name, it needs only two simple steps: **fold the samples into a table, then take an FFT along each row**.
- The simulation checks this recipe against the definition and shows what simple signals (an echo, a tone, a spinning pulse train) look like after the transform.

2 The concept in brief

- **Definition.** For a signal $r[k]$ of MN samples,

$$Y[m,n] = \sum_{q=0}^{N-1} r[m+qM]\, e^{-j2\pi nq/N}.$$

- **Recipe in plain words.**
  1. Cut the sample stream every M samples: this gives N blocks (the Zak periods).
  2. Stack the blocks side by side as the columns of an $M \times N$ table $R[m,q] = r[m+qM]$.
  3. Take an N-point FFT along each row (across q). The result is $Y[m,n]$.
- **Why it makes sense.** Row m of the table holds the sample at the *same position* m in every period. If something sits at that position, but its phase changes from period to period, the FFT over q measures how fast it rotates. That rotation rate is the **Doppler**; the position m is the **delay**.
- **Three building blocks to remember:**

Signal in time	After the Zak transform	Why
Single impulse at (m_0, q_0)	a full **row** (all Doppler bins)	it occurs in one period only, so it is spread equally over all Doppler bins
Complex tone $e^{j2\pi n_0 k/MN}$	a full **column** (all delays)	fixed Doppler, spread over all delays
Pulse train spinning as $e^{j2\pi n_0 q/N}$	a single **point** (m_0, n_0) of height N	one echo with one Doppler

- **Why this matters.** A real channel is a few echoes, each with a delay and a Doppler. In the DD grid each echo is a separate point, even when the received waveform looks like noise.
- **Cost.** Evaluating the sum directly costs about MN^2 operations; the FFT costs about $M\frac{N}{2}\log_2 N$.

- **Scaling.** This definition has no $1/N$ factor, so the transform multiplies the signal energy by N.

One-line memory aid: Zak = reshape into $M \times N$ + FFT along each row. Impulse → row, tone → column, spinning pulse train → point.

3 What was simulated

- The recipe step by step on a random signal (M=16, N=8).
- The three building blocks above on the same grid (M=16, N=8).
- A mixed signal (a row, a column, a point and noise added together) to see whether the DD view separates them.
- The same transform computed three ways (FFT, direct sum, matrix product) and compared, plus the cost of the direct way against the FFT way.

4 Results and discussion

4.1 The recipe: reshape, then FFT

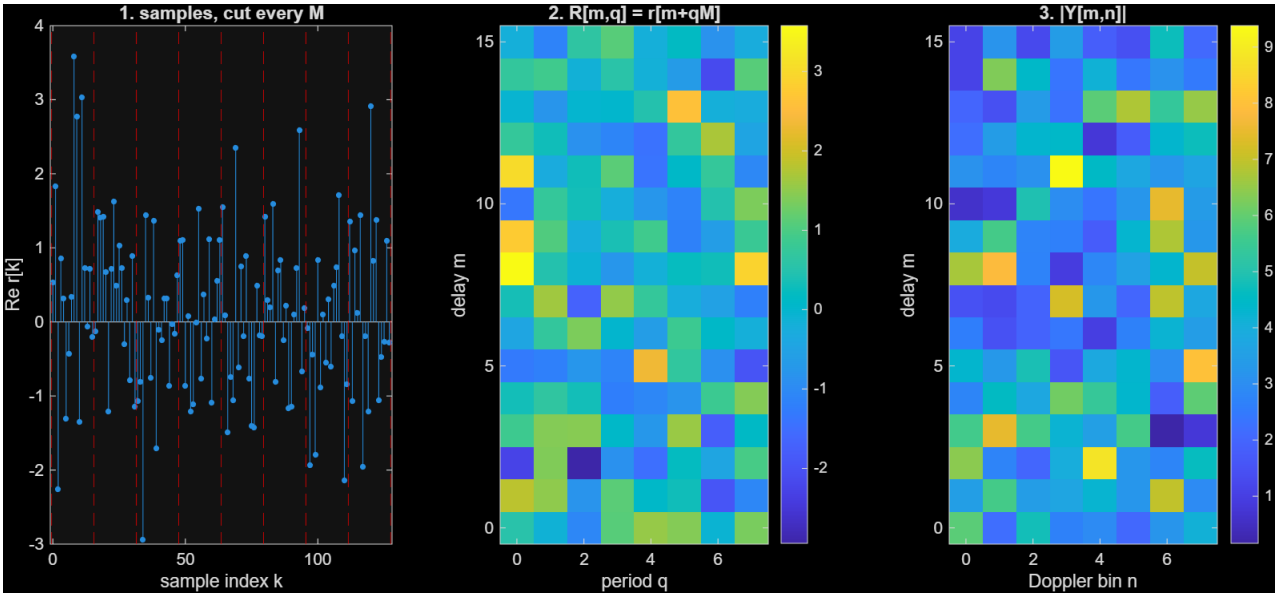


Figure 1: Left: samples (real part) with a cut every M=16 (red lines). Middle: the 16×8 table (real part). Right: $|Y|$ after the row-wise FFT.

- **Expected:** 128 samples cut into 8 blocks of 16 become a 16×8 table; the FFT then turns the 8 columns into 8 Doppler bins.
- **Observed:** the three panels show exactly this chain; a random signal gives a random-looking $|Y|$, as it should.
- Takeaway: nothing mysterious is happening; the Zak transform only reorganises samples and applies an FFT.

4.2 An echo becomes a row

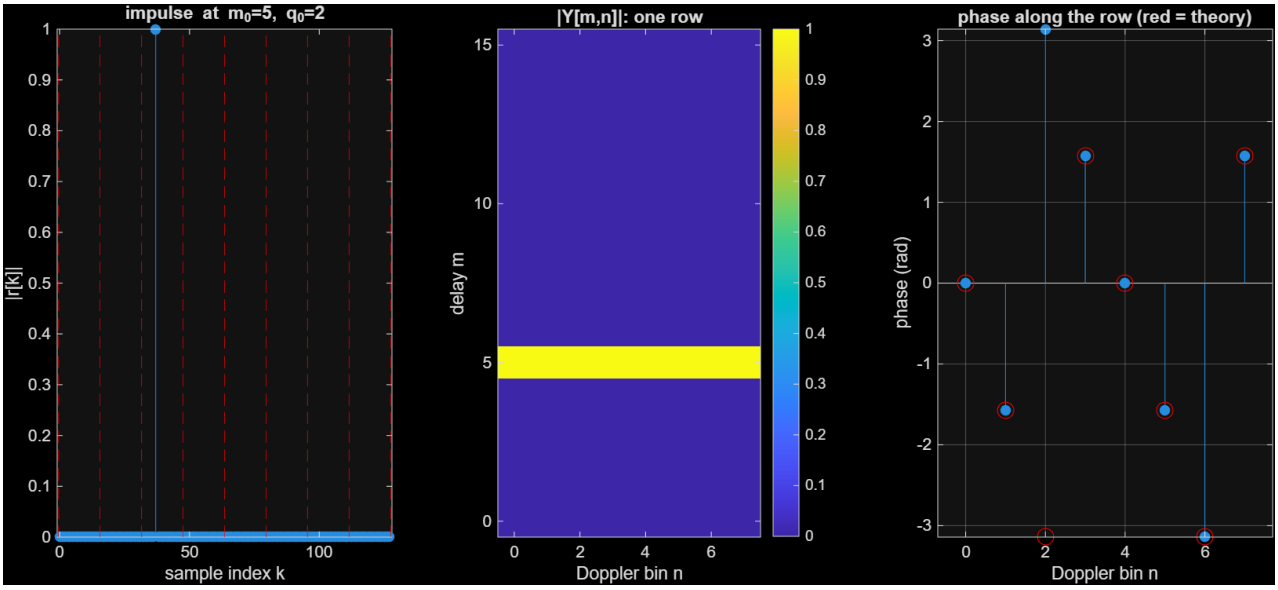


Figure 2: Impulse at m_0=5, q_0=2 (sample $5 + 2 \cdot 16 = 37$). Right: phase along the row (blue) against theory (red circles).

- **Expected:** one row m=5 with magnitude 1 in every Doppler bin, and phase $-2\pi q_0 n/N = -\pi n/2$, i.e. a straight phase ramp.
- **Observed:** the bright stripe is exactly at row 5, and the phase drops by $\pi/2$ per bin (wrapping around at $\pm\pi$), matching the red circles.
- The marker at n=2 is drawn at $+\pi$ while theory is drawn at $-\pi$; these are the same angle, so this is rounding, not an error.
- Takeaway: the delay shows up as the row; the period q_0 of the impulse shows up only as a phase slope.

4.3 A tone becomes a column

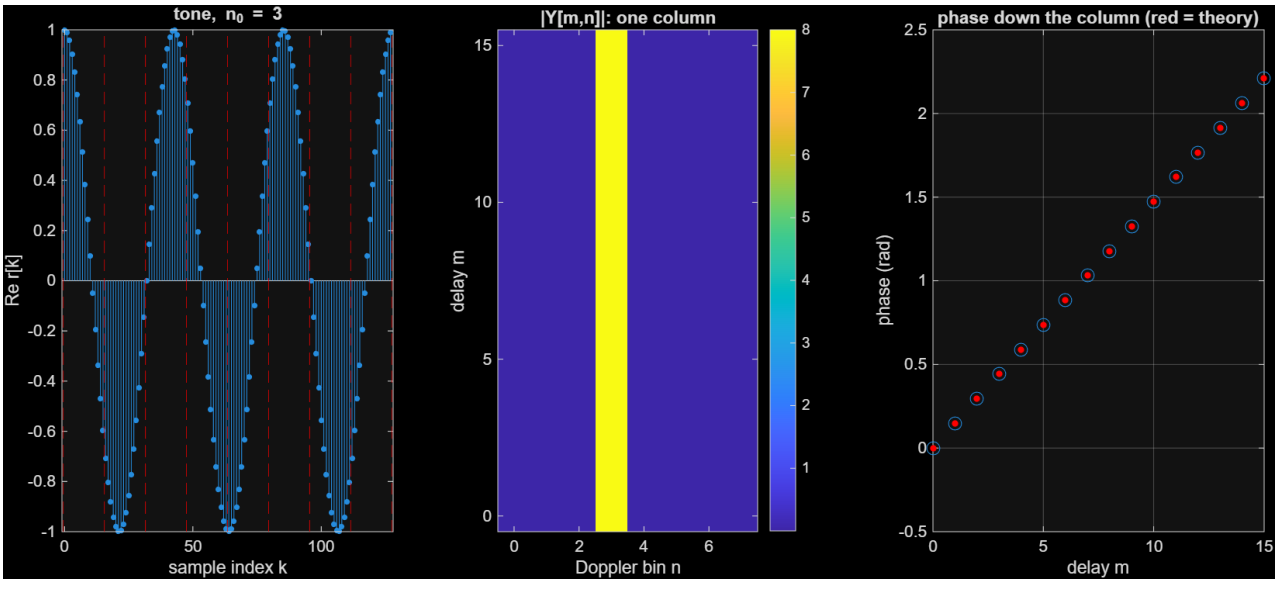


Figure 3: Tone with n_0=3 (three cycles in the frame). Right: phase down the column (blue) against theory (red).

- **Expected:** one column n=3 of height N=8, with phase growing linearly as $2\pi n_0 m/(MN)$.
- **Observed:** the bright stripe is at column 3, with height 8; the phase reaches about 2.2 rad at m=15, which equals $2\pi \cdot 3 \cdot 15/128$.
- Takeaway: a steady tone is the opposite of an echo; it has one fixed Doppler and no particular delay.

4.4 A spinning pulse train becomes a single point

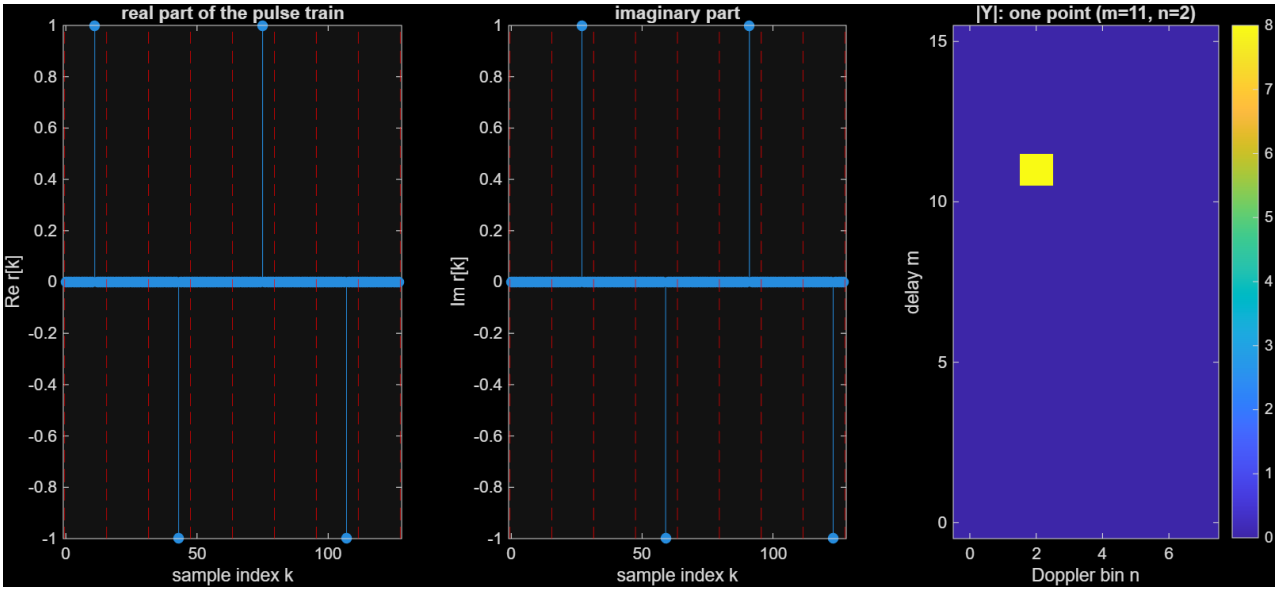


Figure 4: Pulse train with one pulse per period at position m_0=11, rotating as $e^{j2\pi n_0 q/N}$ with n_0=2. Left and middle: real and imaginary parts of the signal. Right: $|Y|$.

- **Expected:** one DD point at $(11, 2)$ with height N=8.
- **Observed:** the $|Y|$ plot shows a single bright cell at $(11, 2)$ with height 8, and zero elsewhere. The pulses are at samples $11, 27, 43, \ldots$ and their phase rotates by 90° from one period to the next (real part $+1, 0, -1, 0$, imaginary part $0, +1, 0, -1$).
- Takeaway: this is the signal that *is* one DD symbol, and it will be the building block of the inverse transform in the next report.

4.5 A noisy-looking signal that is not noise

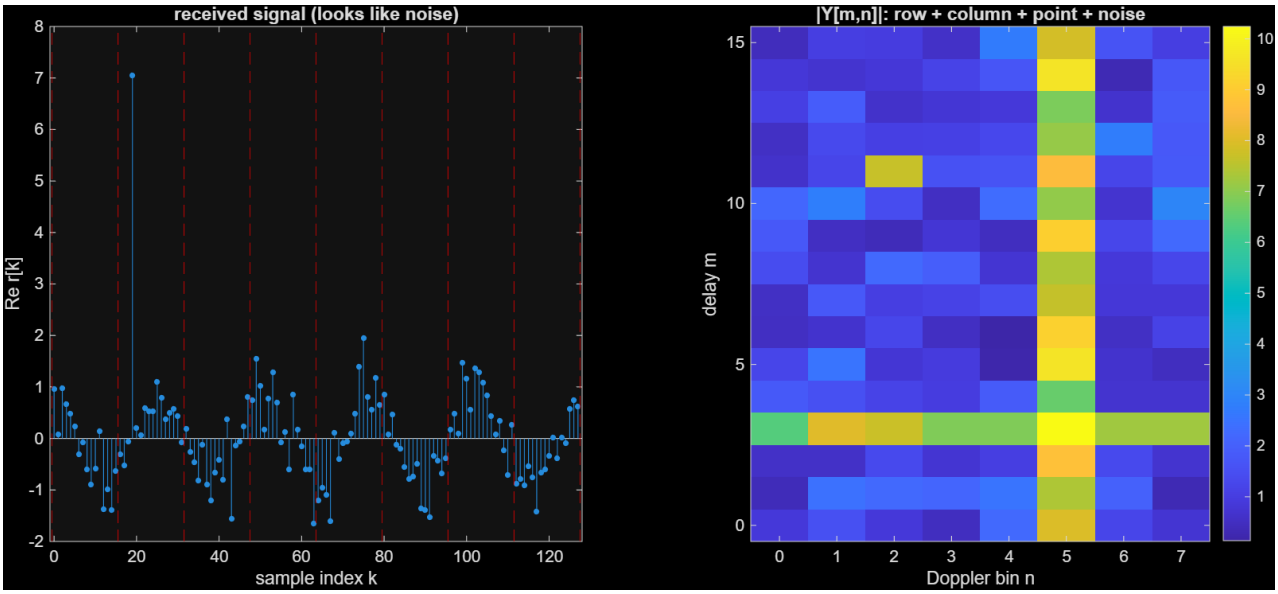


Figure 5: Left: sum of an impulse, a tone, a pulse train and noise in the time domain (real part). Right: its $|Y[m,n]|$.

- **Expected:** the transform is linear, so each ingredient should appear separately in the DD grid.
- **Observed:** in time the signal looks like irregular noise with one large spike. In the DD grid we see a horizontal stripe at m=3, a vertical stripe at n=5, a bright cell at $(11, 2)$, and a low noisy background.
- Takeaway: this is the main reason to use the DD domain: structure that is hidden in time becomes visible as a few simple shapes.

4.6 The three ways agree, and the FFT is cheaper

Table 1: Agreement with theory and between methods.

Check	Maximum difference
FFT vs. direct sum (M=N=64)	$\approx 6 \times 10^{-13}$
FFT vs. matrix product (M=N=64)	$\approx 5 \times 10^{-13}$
Impulse, tone, DD point against their formulas	$\lesssim 10^{-14}$
Energy ratio $\sum \lvert Y\rvert^2 / \sum \lvert r\rvert^2$ (N=64)	$64 = N$

- These differences are floating-point rounding errors accumulated over many operations, so the three methods compute the same transform.
- In the mixed signal the impulse has amplitude N=8 (so that the row is as strong as the others), the tone and the pulse train have amplitude 1, and the noise has $\sigma = 0.5$ per complex sample; in the noise-only bins the expected $\overline{|Y|^2}$ is $N\sigma^2 = 2$.
- The energy ratio equals N, confirming the missing $1/N$ factor noted earlier.

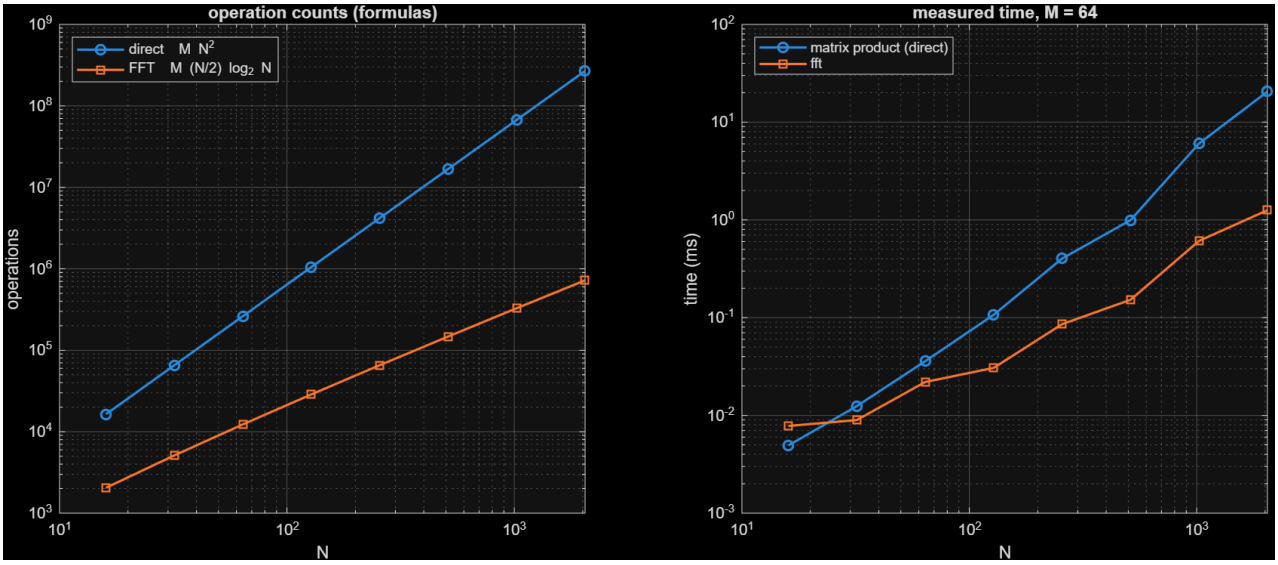


Figure 6: Left: operation counts, MN^2 (direct) against $M\frac{N}{2}\log_2 N$ (FFT), M=64. Right: measured computation time.

- **Expected:** the FFT should win more and more as N grows: for M=64, N=2048 the formulas predict a factor of about 370.
- **Observed:** the measured gap is smaller, about 16× at N=2048 (roughly 20 ms against 1.3 ms), and at the smallest size shown (N=16) the direct product is even slightly faster.
- **Why they differ:** MATLAB's matrix multiplication is heavily optimised and the FFT has fixed overheads, so the formulas count operations, not seconds. The growth trend is the same: the direct curve rises much faster with N.

5 Summary

- Zak = reshape into an $M \times N$ table, then FFT along each row; verified against the definition to $\sim 10^{-13}$.
- An echo gives a row, a tone gives a column, and a spinning pulse train gives a single point of height N; all agree with theory to $\sim 10^{-14}$ or better.
- Structure hidden in a noise-like waveform is clearly visible in the DD grid.
- The FFT route is far cheaper than the direct sum for large N (formula ratio 370 at N=2048; measured 16).

Appendix: details

- Figures come from the MATLAB script `S2_zak_transform.m` (functions `dzt`, `dzt_bruteforce`), using MATLAB's dark theme.
- The error between the direct sum and the FFT grows with the Doppler index n (error map in the script's output), because the direct sum evaluates complex exponentials with larger arguments for larger n; the effect is at the 10^{-13} level.
- The tabulated agreement values were reproduced independently in a second implementation (Python/NumPy); the timing values are read from the plot, so they are approximate and machine dependent.